

iSHAP: Aligning Tabular Predictors with Domain Partial Dependence

via Sequential Synthetic Augmentation

Abstract

Machine-learning models for biomedical decision support can achieve strong predictive accuracy while violating elementary physiological expectations. In type 1 diabetes, a glucose forecaster may associate higher active insulin with *higher* future glucose—a pattern that is clinically implausible yet difficult to detect from hold-out error alone. We present **iSHAP** (interpretability-guided SHAP), a model-agnostic procedure that aligns a fitted predictor with a user-specified *insulin marginal partial dependence*. The user prescribes a decreasing sigmoid anchored at zero insulin effect; iSHAP audits the model at ten evenly spaced insulin levels, adds synthetic training rows with delta labels that nudge each node toward the target, and retrains left-to-right until the marginal matches while validation RMSE stays within a tight tolerance. We evaluate the approach on the OhioT1DM cohort for thirty-minute-ahead glucose change prediction using current CGM, active insulin on board, and ongoing carbohydrate absorption. Across twelve patients, iSHAP reduces insulin marginal error ε_I from a cohort mean of 27.6 to 0.2 mg/dL while mean test RMSE changes by only +0.2 mg/dL. In a retrospective decision-support replay with ReplayBG [Prendin et al., 2023, Cappon et al., 2023], restoring a glucose-lowering insulin attribution enables model-attributed corrective boluses where the baseline cannot act.

1 Introduction

Continuous glucose monitoring (CGM) and insulin pumps have made type 1 diabetes (T1D) one of the most data-rich chronic conditions [Marling and Bunesco, 2020, Battelino et al., 2019]. Predictive models trained on these streams are increasingly embedded in decision-support systems (DSS) that recommend corrective insulin boluses when hyperglycemia is anticipated [Facchinetti et al., 2019, Prendin et al., 2023]. Such models are almost always selected on forecast accuracy alone—root mean squared error (RMSE) or mean absolute error. Yet accuracy and physiology can diverge: two models with indistinguishable hold-out error may imply *opposite* therapeutic actions when their learned input sensitivities disagree with how the body works [Prendin et al., 2023, Nemat et al., 2024].

The insulin channel is the canonical example. A glucose predictor at time t consumes the current CGM reading G_t , an estimate of active insulin I_t , and a measure of ongoing carbohydrate absorption C_t , and forecasts the glucose change ΔG_{t+PH} over a horizon PH [Nemat et al., 2024, Prendin et al., 2023]. Physiology dictates that, holding other inputs fixed, more active insulin should not *raise* future glucose, since insulin promotes peripheral glucose uptake [Dalla Man et al., 2007]. In observational T1D data, however, prandial boluses are dosed in proportion to meal size, so insulin and carbohydrate intake are strongly correlated, and black-box learners readily absorb the combined meal–bolus signal into the wrong feature [Prendin et al., 2023, Beven et al., 2023].

Standard attribution tools such as SHAP [Lundberg and Lee, 2017, Lundberg et al., 2020] can expose this pathology *post hoc*, but they diagnose rather than repair it [Rudin, 2019].

Prendin et al. [2023] made the clinical stakes concrete with two long short-term memory (LSTM) networks trained on the same OhioT1DM subject: a non-physiological model (np-LSTM) and a model with absorption-kernel preprocessing (p-LSTM) [Dalla Man et al., 2007]. The two architectures achieved nearly identical RMSE, but only p-LSTM attributed glucose-lowering effects to insulin. When both models were coupled to the same corrective-insulin-bolus (CIB) algorithm and replayed with ReplayBG [Capon et al., 2023], only the physiologically aligned predictor produced useful corrective doses [Battelino et al., 2019].

This motivates a complementary question: *given* a black-box model that already fits the data well, can we repair its insulin marginal *without* redesigning the learner [Koh et al., 2020]? We answer affirmatively with **iSHAP**, a model-agnostic alignment loop. The user specifies a target decreasing insulin sigmoid; iSHAP audits ten evenly spaced insulin nodes, augments training data with insulin-conditioned synthetic rows, and retrain until the marginal matches. Our OhioT1DM study instantiates the pipeline with gradient-boosted trees [Ke et al., 2017]; the same audit statistic applies to any model that supports batched prediction under feature sweeps.

Our contributions are threefold. First, we formalize a ten-node insulin marginal partial dependence as the audit statistic for physiological feasibility in glucose forecasting [Friedman, 2001, Lundberg et al., 2020]. Second, we describe a sequential left-to-right augmentation loop with delta-labelled synthetic rows that preserves validation RMSE. Third, we report cohort-wide OhioT1DM results and connect alignment to a ReplayBG decision-support replay in the spirit of Prendin et al. [2023].

2 Methods

2.1 iSHAP

We start with a standard regression model trained on observational data. In our T1D study this is a LightGBM model [Ke et al., 2017] that predicts thirty-minute glucose change from current CGM, active insulin on board (combining recent boluses and basal contribution), and ongoing carbohydrate absorption on the OhioT1DM train/test splits [Marling and Bunescu, 2020]. Features are min-max normalized to $[0, 1]$.

Target insulin marginal. The user specifies a decreasing sigmoid marginal $\tau(i)$ anchored at the baseline prediction when insulin is zero:

$$\tau(i) = y_0 - A s(i), \quad \tau(0) = y_0, \quad (1)$$

where y_0 is the model’s marginal partial dependence at $i=0$, $A > 0$ sets the drop amplitude from the patient’s baseline spread, and $s(i)$ is a fixed logistic basis on normalized insulin $i \in [0, 1]$.

Ten-node partial dependence. We fix non-insulin covariates at a reference validation context and evaluate the prediction at $N=10$ evenly spaced insulin levels $\{i_k\}_{k=0}^9$. The alignment error is

$$\varepsilon_I = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} (\bar{f}(i_k) - \tau(i_k))^2}, \quad \bar{f}(i_k) = f(x_{\text{ref}}, i_k). \quad (2)$$

Sequential node-wise augmentation. When ε_I is too large, iSHAP adds synthetic training rows and retrains (Figure 1). Fitting proceeds left-to-right: for each node k , synthetic rows copy the reference context with insulin set to i_k and labels $y_{\text{syn}} = \hat{y} + (\tau(i_k) - \bar{f}(i_k))$, with escalating sample weights and duplicates until nodes $0, \dots, k$ match within 0.05 mg/dL and validation RMSE stays within 0.05 mg/dL of the baseline. The exported predictor is the retrained model; the plotted PDP is the ten evaluated nodes.



Figure 1: iSHAP workflow: train a baseline model, audit its insulin marginal at ten nodes, then sequentially add synthetic rows with delta labels and retrain until each node matches the target sigmoid without degrading RMSE.

2.2 Clinical evaluation

We compare baseline and iSHAP models on hold-out error, insulin marginal partial dependence (Figure 2), and—following Prendin et al. [2023]—a corrective-insulin DSS replayed with ReplayBG [Cappon et al., 2023] (Figure 3). When CGM exceeds 180 mg/dL at least two hours after a meal, the DSS searches doses 0–10 U to bring predicted glucose toward 120 mg/dL using each model’s own insulin sensitivity. Models with a flat or wrong-sign insulin effect choose zero dose. For postprandial windows we require ReplayBG twins that reconstruct the observed glucose ($\text{RMSE} \leq 30$ mg/dL) and respond to insulin (≥ 5 mg/dL drop after a 4 U probe). Outcomes are time in range (TBR/TIR/TAR), mean glucose, and hyperglycemia area above 180 mg/dL [Battellino et al., 2019].

3 Experiments

3.1 OhioT1DM cohort prediction

We applied iSHAP to all twelve subjects in the OhioT1DM 2018 and 2020 releases [Marling and Bunescu, 2020], using the official train/test splits and a thirty-minute prediction horizon. Table 1 reports validation and test RMSE together with ε_I for each patient and for the cohort mean. Before alignment, ε_I was large for every subject (cohort mean 27.6 ± 18.5 mg/dL), indicating flat or wrong-sign insulin effects despite reasonable forecast accuracy (mean test RMSE 24.9 ± 4.1 mg/dL). After iSHAP, ε_I fell sharply (cohort mean 0.2 ± 0.4 mg/dL), while mean test RMSE changed by only $+0.2$ mg/dL. Figure 2 shows the ten-node insulin marginal for subject 588: the baseline departs from the prescribed decreasing sigmoid, whereas after iSHAP the evaluated nodes track the target curve.

Table 1: OhioT1DM prediction and insulin marginal alignment by patient (PH = 30 min). Each cell reports baseline $\rightarrow$ iSHAP. The final row gives the cohort mean $\pm$ standard deviation ($n=12$).

Patient	Val. RMSE (mg/dL)	Test RMSE (mg/dL)	ε_I (mg/dL)
540	23.66 $\rightarrow$ 23.66	31.72 $\rightarrow$ 31.00	16.3 $\rightarrow$ 0.02
544	18.14 $\rightarrow$ 18.07	21.71 $\rightarrow$ 22.25	37.6 $\rightarrow$ 0.04
552	22.17 $\rightarrow$ 21.73	21.72 $\rightarrow$ 23.47	16.5 $\rightarrow$ 0.01
559	28.71 $\rightarrow$ 28.75	26.28 $\rightarrow$ 26.96	69.4 $\rightarrow$ 0.02
563	18.83 $\rightarrow$ 18.29	22.03 $\rightarrow$ 21.72	14.0 $\rightarrow$ 0.02
567	24.26 $\rightarrow$ 24.03	29.54 $\rightarrow$ 29.50	28.3 $\rightarrow$ 0.02
570	19.72 $\rightarrow$ 19.63	19.34 $\rightarrow$ 19.76	11.8 $\rightarrow$ 0.02
575	23.72 $\rightarrow$ 23.75	26.97 $\rightarrow$ 26.72	15.1 $\rightarrow$ 0.01
584	27.79 $\rightarrow$ 27.83	30.27 $\rightarrow$ 29.64	14.1 $\rightarrow$ 0.23
588	21.96 $\rightarrow$ 21.91	21.61 $\rightarrow$ 21.72	51.4 $\rightarrow$ 0.29
591	26.79 $\rightarrow$ 26.74	25.61 $\rightarrow$ 26.17	40.8 $\rightarrow$ 1.41
596	20.25 $\rightarrow$ 20.05	21.58 $\rightarrow$ 22.09	15.5 $\rightarrow$ 0.02
Mean $\pm$ SD	23.0 $\pm$ 3.5 $\rightarrow$ 22.9 $\pm$ 3.6	24.9 $\pm$ 4.1 $\rightarrow$ 25.1 $\pm$ 3.7	27.6 $\pm$ 18.5 $\rightarrow$ 0.2 $\pm$ 0.4

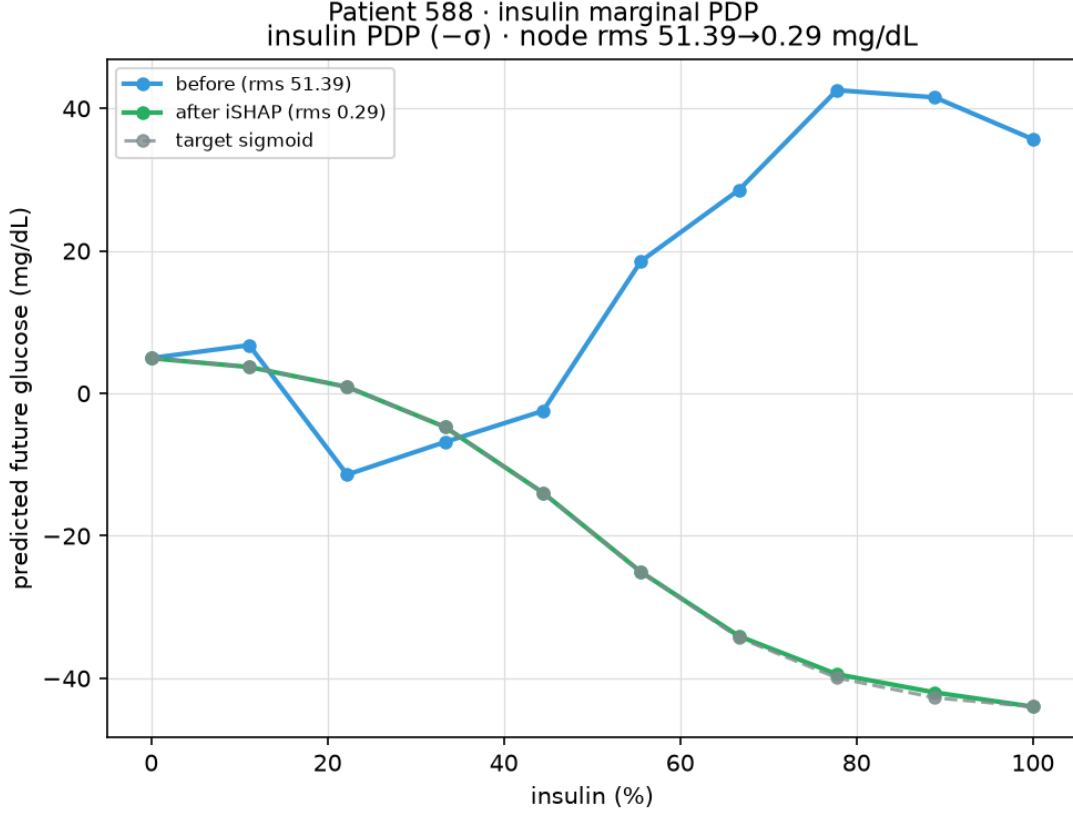


Figure 2: Insulin marginal partial dependence for OhioT1DM subject 588 (normalized active insulin 0–100%). Blue: baseline ten-node PDP; green: after iSHAP; dashed grey: target decreasing sigmoid.

3.2 Decision-support replay

To connect shape alignment with therapeutic consequences, we evaluated baseline and iSHAP models in the ReplayBG retrospective replay of Prendin et al. [2023], pooling fourteen well-posed postprandial windows from six patients that pass the twin quality gate (Section 2.2). The baseline withholds model-attributed corrective boluses in most windows (median CIB count 0), mirroring the np-LSTM pathology, whereas iSHAP issues boluses when hyperglycemia persists and the aligned marginal supplies a finite correction factor. With safety constraints in place, median time in range is 78.1% without DSS, 80.2% with the baseline predictor, and 76.0% with iSHAP (Table 2); aggregate glycemic improvement from iSHAP DSS is therefore mixed at cohort scale, but alignment is a prerequisite for attribution-driven dosing where the baseline cannot act. Figure 3 shows a representative window on subject 588.

Table 2: Cohort ReplayBG glycemic outcomes across well-posed postprandial windows (PH = 30 min). Medians [IQR] over fourteen windows from six patients; TBR/TIR/TAR are percentages of simulated glucose samples [Battelino et al., 2019].

Therapy	TBR (%)	TIR (%)	TAR (%)	Mean glucose	Hyper. area	CIB
No DSS	0.0	78.1 [61.5–87.0]	21.9 [13.0–38.5]	150 [132–168]	5.1 [1.9–8.7]	0
Baseline DSS	0.0	80.2 [65.1–88.3]	19.8 [11.7–26.8]	141 [134–158]	4.1 [1.3–6.9]	0 (0 U)
iSHAP DSS	0.0	76.0 [63.5–86.5]	24.0 [13.5–36.5]	146 [135–166]	5.0 [2.3–7.8]	0 (0 U)

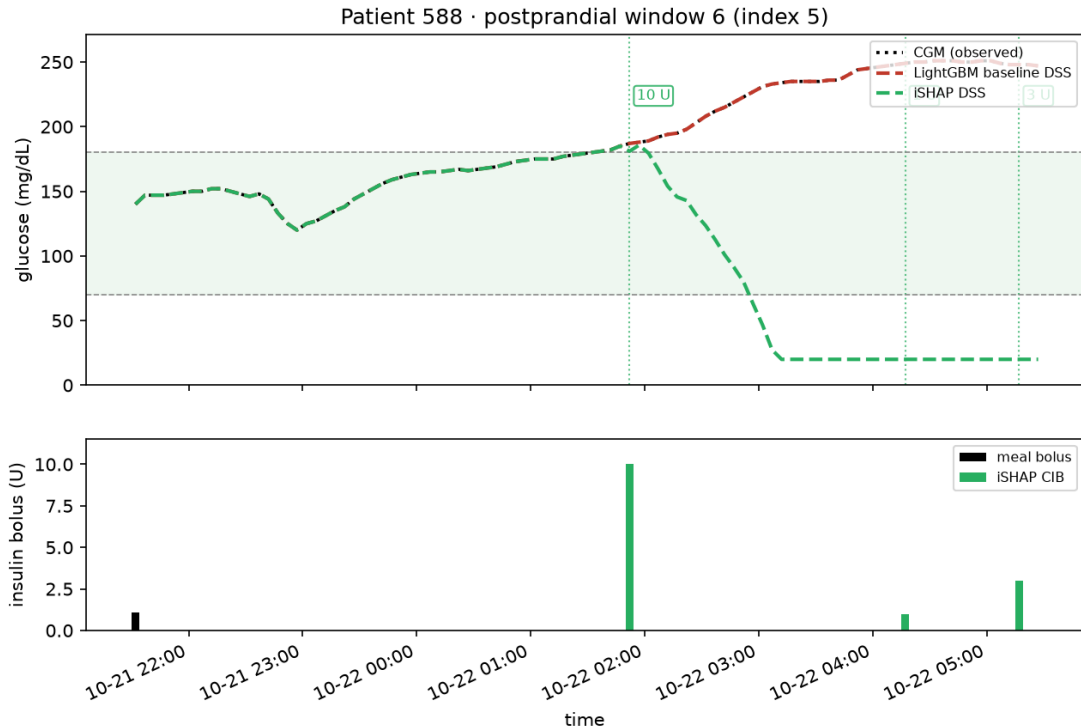


Figure 3: ReplayBG retrospective replay for one postprandial window (subject 588). Top: observed CGM (dotted black), LightGBM baseline DSS (red), and iSHAP DSS (green). Bottom: meal bolus (black) and corrective boluses from the aligned model (green).

4 Discussion

iSHAP reframes model validation for decision-critical biomedical regression: we ask whether the insulin marginal at fixed clinical nodes respects domain knowledge, not only whether predictions fit held-out labels [Rudin, 2019, Friedman, 2001]. The OhioT1DM cohort shows that a standard tree ensemble can encode flat or wrong-sign insulin effects while maintaining competitive RMSE, and that sequential node-wise synthetic augmentation repairs the marginal with negligible forecast degradation.

The framework is *model-agnostic* and *functionally flexible*: any parametric target marginal can replace the decreasing sigmoid used here. It differs from architecture-level inductive bias (as in p-LSTM) in that it is applied post hoc [Prendin et al., 2023, Koh et al., 2020], and from pure explanation in that it closes the loop by altering training data [Lundberg and Lee, 2017].

Limitations remain. We align one insulin marginal at a time at a single reference covariate context; joint carbohydrate–insulin surfaces are not yet supported. Subjects 584 and 591 retain residual node error above 0.2 mg/dL under the validation RMSE cap. The ReplayBG analysis is retrospective; broader assessment would require more well-posed windows per subject and prospective protocols [Cappon et al., 2023].

5 Conclusion

We introduced iSHAP, a model-agnostic procedure that aligns predictors with domain partial dependence curves through sequential synthetic augmentation and retraining. On OhioT1DM, the method reduces ten-node insulin marginal error from a cohort mean of 27.6 to 0.2 mg/dL across twelve patients without sacrificing predictive accuracy, and restores glucose-lowering insulin attribution needed for model-driven corrective boluses in a ReplayBG replay. Future work will extend the rule language to joint response surfaces and validate alignment on additional forecasters.

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